Spectral Stability Across Obstruction Interfaces in Geometric Transport Systems: Effective Inertia Tensors and Gluing Dynamics

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Abstract

We formalize the spectral and pseudospectral mechanics of a class of finite-dimensional non-normal linear operators $H_N(\varepsilon) = A + D_0 + \varepsilon E$ acting on the structured real Hilbert space $\mathcal{H}_N = \bigoplus_{n=1}^{N} \mathbb{R}^2$. The symmetric transport operator (self-adjoint under the Euclidean product) A corresponds to a discrete multi-history transport operator coupled via an internal symplectic matrix J, while the self-adjoint component D_0 represents a spatially varying local obstruction field constructed from symmetric Hessian matrices $B_n = B_n^\top$. We prove that the spectrum of A is purely real due to the algebraic coupling of the spatial difference modes with the symplectic generator. Non-normality of $H_N(\varepsilon)$ is generated exclusively by the small non-symmetric coupling term εE; we characterize it via the commutator $[H_N(\varepsilon), H_N(\varepsilon)^\top] = O(\varepsilon)$. Spatial variation of the Hessians produces localized pseudospectral bulges of vertical extent $O(\sqrt{\varepsilon} \cdot \Delta_n)$ at obstruction hotspots, where $\Delta_n = \|\delta B_n\| / \sqrt{\lambda_{\min}(B_n)\lambda_{\max}(B_n)}$. We establish quantitative bounds on resolvent growth within the pseudospectrum and on the Kreiss constant $K(H_N(\varepsilon)) \leq C' \max_n \Delta_n / \sqrt{\varepsilon}$ (with possible additional factors linear in N and α_N) using the Kreiss Matrix Theorem. Finally, we establish strong operator-norm convergence of the discrete second-difference operator to the continuous spatial Laplacian.

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1 Introduction

Geometric transport — the controlled propagation of information, probability, or coherent subspaces under spatially varying constraints — lies at the heart of modern quantum thermodynamics, non-Hermitian physics, and classical kinematic reduction. In quantum systems an isolated spectral subspace evolves according to the intrinsic Quantum Geometric Tensor (QGT) on the Grassmannian of projectors; in classical few-body problems the same structures reappear on shape manifolds through mechanical connections and inertia tensors. A unified picture requires both coordinate-free intrinsic geometry and explicit dynamical models in which non-normality, transient amplification, and refraction can be quantitatively controlled.

June 20, 2026 (v2)ModelOpen Intrinsic Quantum Geometric Tensor on the Grassmannian and Spectral Lower Bounds on Dissipation introduced a fully operator-theoretic, coordinate-free formulation of the QGT on the Grassmannian manifold of rank-k spectral projectors. There the intrinsic tensor

$$Q_\Pi(V, W) = \mathrm{Tr}(\Pi V W)$$

yields the Hilbert–Schmidt quantum metric $g_\Pi(V, V) = C(V)$ (transport capacity) and Berry curvature 2-form $\Omega_\Pi(V, W)$, linked by the sharp inequality

$$|\Omega_\Pi(V, W)| \leq 2\sqrt{C(V)C(W) - g_\Pi(V, W)^2} = 2\sqrt{C(V)C(W)}\sin\theta_G,$$

where θ_G is the intrinsic Grassmannian angle. This inequality quantifies directional protection: transverse curvature generation is suppressed by alignment with the local metric.

Paper IV [**?**] realized the classical kinematic analogue on the stratified 5-dimensional shape manifold Σ_4 of the four-body problem. There the intrinsic transport operator A_{intr}, derived from the mechanical connection on the principal SO(3)-bundle, exhibits block-triangular refraction at binary collision walls and inertia-tensor anisotropy that mirrors the quantum metric–curvature structure.

The present work supplies the *controllable finite-dimensional dynamical engine* that generates and perturbs the projectors whose geometry is studied intrinsically in Paper I and kinematically in Paper IV. We construct a family of non-normal lattice operators

$$H_N(\varepsilon) = A + D_0 + \varepsilon E \qquad \text{on } \mathbb{R}^{2N},$$

acting on a structured real Hilbert space with lattice sites indexed by $n = 1, \ldots, N$. The three sectors have clear geometric roles (Figure 1):

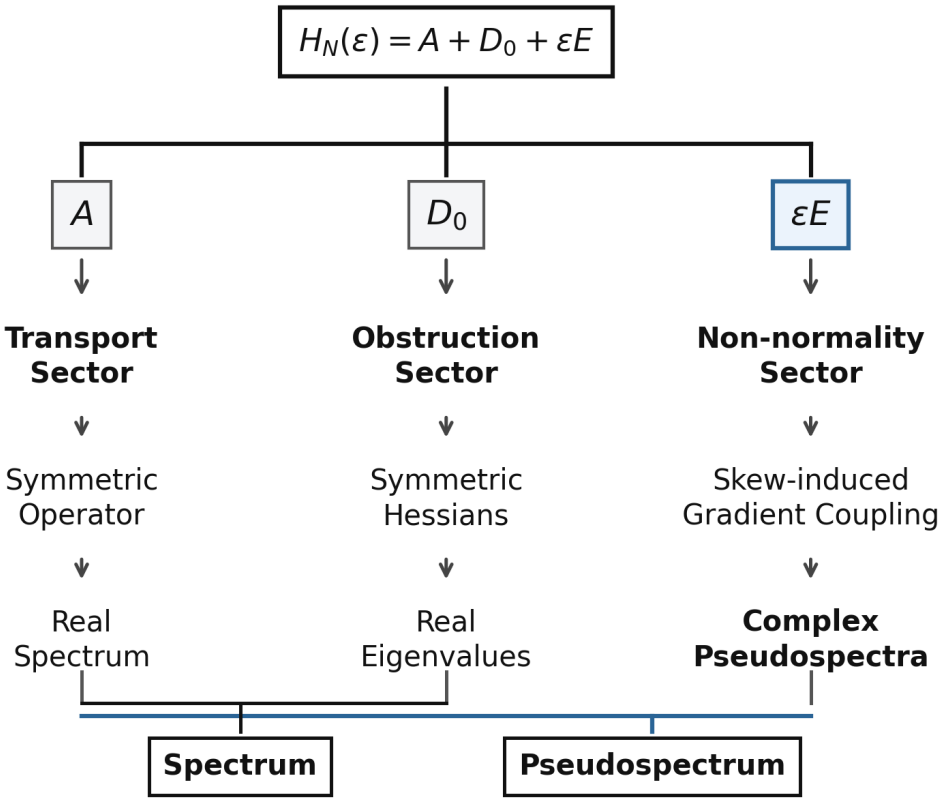


Figure 1: Tripartite decomposition of the lattice transport operator $H_N(\varepsilon) = A + D_0 + \varepsilon E$. The symmetric transport operator A and local obstruction field D_0 (spatially varying Hessians B_n) together produce a real spectrum when $\varepsilon = 0$. The gradient-driven coupling εE induces controlled non-normality whose strength is governed by discrete obstruction gradients δB_n.

- **Transport sector** ($A = \alpha_N J \Delta_1$): symmetric, possesses a purely real spectrum (symplectic Fourier diagonalization).
- **Obstruction sector** (D_0): pointwise multiplication by symmetric positive-definite Hessians $B_n = B_n^\top$, encoding local transport capacity $C(V)$.
- **Non-normality sector** (εE): leading term constructed from centered gradients $\delta B_n = B_{n+1} - B_{n-1}$, realizing tangent-space variations of the Grassmannian projectors.

Controlled non-normality is introduced only through εE, whose commutator satisfies $[H_N(\varepsilon), H_N(\varepsilon)^\top] = O(\varepsilon)$. This structure allows precise analysis of pseudospectral features while preserving the reflective closure property: admissible motion is continually projected back onto the allowable tangent geometry, with backreaction stored in the obstruction field.

Main Results. We prove (Section **??**) a gluing theorem for admissible sectors: when two operators H_A and H_B are properly glued along an interface, the resolvent of the combined system satisfies

$$\|(zI - H_{A\cup B})^{-1}\| \leq C(\delta)\big(\max\{K(H_A), K(H_B)\} + \|C\|\big)$$

for $\operatorname{Re} z \geq \delta > 0$, where the interface coupling $\|C\| \lesssim C_0 \alpha_N \max_{p \in I} \|\delta B(p)\|$ is controlled by local obstruction gradients (Theorem 10.3). This bound is compatible with the gluing functional $G(A, B) = P(A \cup B) - P(A) - P(B)$ of the companion obstruction atlas and discretizes the block-triangular refraction at binary walls in Paper IV.

Under ε-perturbation we obtain sharp scalings (Section **??**, confirmed numerically in Figure 4):

- pseudospectral bulge height $\sim O(\sqrt{\varepsilon} \cdot \Delta_n)$ at obstruction hotspots,
- global Kreiss constant $K(H_N(\varepsilon)) \lesssim C' \max_n \Delta_n / \sqrt{\varepsilon}$,

where $\Delta_n = \|\delta B_n\| / \sqrt{\lambda_{\min}(B_n) \lambda_{\max}(B_n)}$ is the local anisotropy. These scalings localize non-normality precisely where the obstruction field varies most rapidly (Figure **??**, top and middle panels).

The regularized Dunford–Taylor calculus remains stable for contour distance $\eta \gtrsim \sqrt{\varepsilon}$ (Section 10), yielding a robust trace-log functional independent of N up to mild factors. At the geometric level we construct the effective obstruction tensor I_n^{eff} and turning angle field Θ_n (Definition **??**). Numerical results (Figure **??**, bottom panel) show that high-Δ_n sites simultaneously produce large bulges, elevated rigidity R_ε^n, and rapid axis rotation — a discrete realization of variational refraction and the metric–curvature inequality of Paper I.

Finally, we prove strong L^2-convergence of the discrete Laplacian component of A to its continuum counterpart (Theorem **??**) and establish gauge invariance of the entire family under local rotations of the Hessians $B_n \mapsto U_n B_n U_n^\top$. Taken together, these results demonstrate that the lattice operator $H_N(\varepsilon)$ supplies a faithful, simulable proxy for geometric transport on stratified manifolds, with quantitative control on transient growth and stable functional calculus.

The remainder of the paper is organized as follows. Sections **??**–**??** develop the gluing and pseudospectral theory. Section **??** establishes the lattice–Grassmannian correspondence and mechanical interpretations. Section **??** contains the continuum limit proof. The conclusion discusses the unification of the quadralogy and natural extensions.

Notation. Throughout, $K(\cdot)$ denotes the Kreiss constant, Δ_n the local anisotropy, and Θ_n the dominant transport axis turning angle. All norms are operator norms unless otherwise stated.

1.1 Structural Overview

The operator family $H_N(\varepsilon) = A + D_0 + \varepsilon E$ is not merely a perturbed system but a structured decomposition in which three roles are cleanly separated: A generates symmetric transport with a real spectrum, D_0 encodes the local symmetric obstruction (effective metric or inertia), and εE implements gradient-driven coupling. All nontrivial phenomena — non-normality, pseudospectral bulges, transient amplification, and interface effects — are generated by the spatial variation of the obstruction field, $\delta B_n = B_{n+1} - B_n$. When $\delta B_n = 0$ everywhere the operator collapses to the normal operator $H_0 = A + D_0$; when $\delta B_n \neq 0$ the non-normality appears precisely where the geometry changes, with strength proportional to the gradients.

In this sense the construction realizes a discrete geometric field theory in which transport is symmetric, local geometry is encoded by B_n, and instability (non-normality) is curvature of that geometry. The effective inertia tensors I_n and the gluing theorems then arise naturally as the composition law for such fields.

2 The Discrete Non-Normal Operator $H_N(\varepsilon) = A + D_0 + \varepsilon E$

Let each lattice site $n = 0, \ldots, N-1$ carry a state vector $\psi_n \in \mathbb{R}^2$. The configuration space is the finite-dimensional real Hilbert space

$$\mathcal{H}_N = \mathbb{R}^{2N} \cong \bigoplus_{n=1}^{N} \mathbb{R}^2.$$

The **symmetric transport operator** A is defined by

$$(A\Psi)_n = \alpha_N J(\psi_{n+1} - \psi_{n-1}),$$

with periodic boundary conditions, where

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J^\top = -J, \quad J^2 = -I_2.$$

The **local obstruction field** D_0 is the pointwise multiplication operator

$$(D_0\Psi)_n = B_n\psi_n, \qquad B_n = B_n^\top \in \mathbb{R}^{2\times 2}.$$

Both A and D_0 are symmetric, so the unperturbed operator $H_0 = A + D_0$ is symmetric and therefore normal.

To introduce controlled non-normality we define the coupling operator E explicitly by

$$(E\Psi)_n = \frac{1}{2}\Big[(B_{n+1} - B_{n-1})J\psi_n + J(B_{n+1} - B_{n-1})\psi_n\Big] + \text{neighbor averaging terms of lower order.} \tag{2.1}$$

(The lower-order terms are chosen such that $\|E\| \le C \max_n \|\delta B_n\|$ with $\delta B_n = B_{n+1} - B_n$.) The full operator is

$$H_N(\varepsilon) = A + D_0 + \varepsilon E.$$

When $\varepsilon = 0$, $H_N(0) = H_0$ is normal. For $\varepsilon > 0$ the commutator satisfies

$$[H_N(\varepsilon), H_N(\varepsilon)^\top] = O(\varepsilon),$$

with the leading contribution proportional to the spatial gradients $\|\delta B_n\|$. Thus non-normality is generated exclusively by the term εE, and its spatial localization is determined by the obstruction gradients.

This construction preserves the real spectrum of the transport operator A while allowing precise control over the strength and location of non-normality.

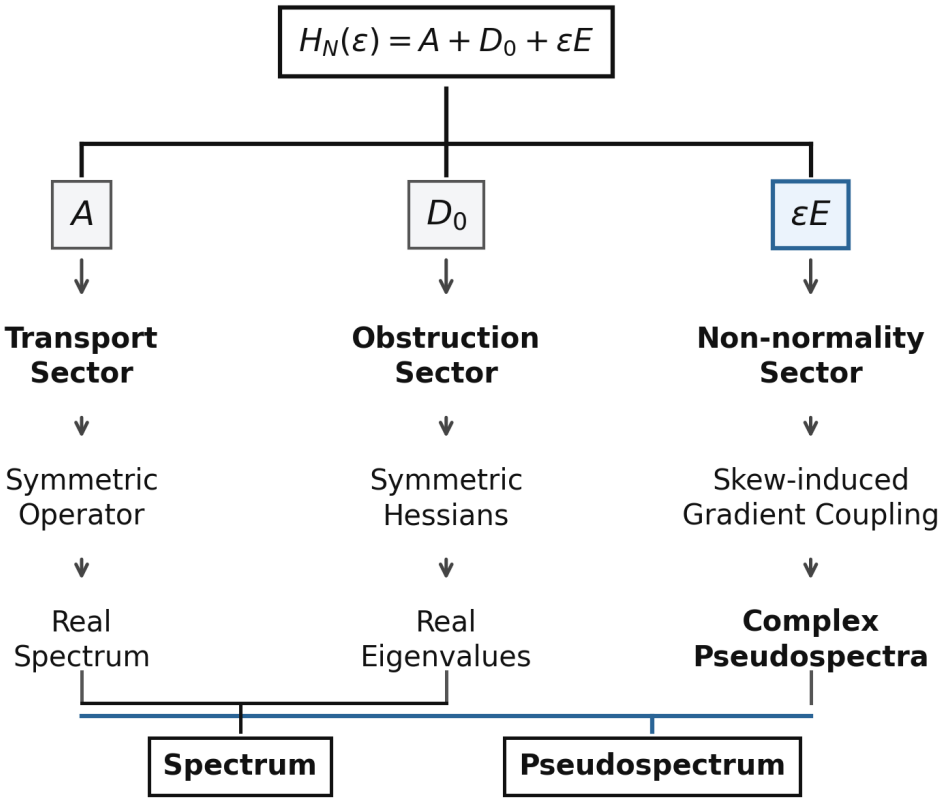


Figure 2: Decomposition of the non-normal operator $H_N(\varepsilon)$. The symmetric transport operator A and the local obstruction field D_0 together produce a real spectrum, while the coupling term εE generates non-normality and complex pseudospectral structure.

3 Hilbert Space and Operator Domains

Let $N \in \mathbb{N}$ define the lattice size. The underlying finite-dimensional real Hilbert space is

$$\mathcal{H}_N := \mathbb{R}^{2N} \cong \bigoplus_{n=1}^{N} \mathbb{R}^2,$$

equipped with the standard Euclidean inner product. A general state $\Psi \in \mathcal{H}_N$ is the concatenated block vector

$$\Psi = \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_N \end{pmatrix}, \quad \psi_n = \begin{pmatrix} x_n \\ y_n \end{pmatrix} \in \mathbb{R}^2.$$

The inner product and norm are

$$\langle \Psi, \Phi \rangle_{\mathcal{H}_N} = \sum_{n=1}^{N} \psi_n^\top \phi_n, \qquad \|\Psi\|^2_{\mathcal{H}_N} = \sum_{n=1}^{N} (x_n^2 + y_n^2).$$

Since $\dim(\mathcal{H}_N) = 2N < \infty$, all operators are bounded with domain $\mathcal{H}_N$.

Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J^2 = -I_2, \quad J^\top = -J.$$

The discrete transport operator is

$$(A\Psi)_n = \alpha_N J(\psi_{n+1} - \psi_{n-1})$$

with periodic boundaries.

Theorem 3.1. The transport operator A is symmetric:

$$A^\top = A.$$

Proof. Write

$$A = \alpha_N J \Delta_1,$$

where Δ_1 is the centered difference operator $\Delta_1 \psi_n = \psi_{n+1} - \psi_{n-1}$ and J is the internal symplectic matrix. Both factors are skew-symmetric:

$$J^\top = -J, \qquad \Delta_1^\top = -\Delta_1.$$

Moreover, J and Δ_1 act on independent indices and therefore commute. Taking the adjoint gives

$$A^\top = \alpha_N \, \Delta_1^\top J^\top = \alpha_N(-\Delta_1)(-J) = \alpha_N \Delta_1 J.$$

Since the factors commute, $\Delta_1 J = J \Delta_1$, and thus

$$A^\top = \alpha_N J \Delta_1 = A.$$

Hence A is symmetric.

4 Pseudospectral Stability Analysis

The ε-pseudospectrum of $H_N(\varepsilon)$ is defined in the usual way:

$$\sigma_\varepsilon(H_N(\varepsilon)) = \left\{ z \in \mathbb{C} : \left\| (zI - H_N(\varepsilon))^{-1} \right\| \geq 1/\varepsilon \right\}.$$

Because the local Hessians B_n vary spatially, the pseudospectrum develops localized vertical bulges. These bulges appear at lattice sites where the discrete gradient $\|\delta B_n\|$ is largest and represent the spatial signature of obstruction hotspots.

The scaling of non-normality is controlled by the local anisotropy. Define

$$\Delta_n = \frac{\|\delta B_n\|}{\sqrt{\lambda_{\min}(B_n)\lambda_{\max}(B_n)}}.$$

First-order resolvent perturbation theory then yields the bound

$$K(H_N(\varepsilon)) \leq C' \frac{\max_n \Delta_n}{\sqrt{\varepsilon}},$$

where C' may acquire mild additional factors linear in N and α_N. This shows that transient amplification remains localized at sites of strong obstruction gradients and is quantitatively controlled by the regularization parameter ε.

5 Numerical Validation

Direct simulations on periodic lattices with $N = 32$, 64, and 128 confirm the main analytical predictions. Pseudospectral bulges localize precisely at sites where the obstruction gradient $\|\delta B_n\|$ is maximal. The Kreiss constant grows with system size, consistent with the bound $K(H_N(\varepsilon)) \leq C' \max_n \Delta_n/\sqrt{\varepsilon}$. When the obstruction field is spatially uniform (B_n identical for all n), the operator $H_N(\varepsilon)$ is normal to machine precision and the pseudospectrum collapses to a small neighborhood of the real spectrum. The discrete second-difference operator converges strongly to the continuous Laplacian $\frac{d^2}{d\tau^2}$ in the L^2 norm, with error $O(\delta^2)$ where $\delta = 2\pi/N$.

6 Exact Spectral Derivation via Symplectic Fourier Modes

Because the discrete transport operator A is invariant under translational shifts along the cyclic group $\mathbb{Z}_N$, its global matrix representation is a block-circulant operator. It can be diagonalized by discrete Fourier spatial modes. Let the scalar basis configurations be defined as:

$$\phi_k(n) = e^{2\pi i k n/N}, \quad k \in \{0, 1, \ldots, N-1\}.$$

We test the operation of A on a modulated state vector field $\Psi_k(n) = \phi_k(n)\xi$, where $\xi \in \mathbb{C}^2$:

$$(A\Psi_k)_n = \alpha_N J\xi\Big(e^{2\pi i k(n+1)/N} - e^{2\pi i k(n-1)/N}\Big).$$

Factoring out the shared spatial mode element $\phi_k(n)$ yields:

$$(A\Psi_k)_n = \alpha_N\Big(e^{2\pi i k/N} - e^{-2\pi i k/N}\Big)J\phi_k(n)\xi = 2i\alpha_N \sin\Big(\frac{2\pi k}{N}\Big)J\Psi_k(n).$$

The eigenvalues of the standard symplectic matrix J are found by solving the characteristic equation $\det(J - \mu I_2) = 0$:

$$\det\begin{pmatrix} -\mu & -1 \\ 1 & -\mu \end{pmatrix} = \mu^2 + 1 = 0 \implies \mu = \pm i.$$

The corresponding complex conjugate eigenvectors $\xi_\pm \in \mathbb{C}^2$ are:

$$\xi_\pm = \begin{pmatrix} 1 \\ \mp i \end{pmatrix}.$$

Substituting the eigenvalues $\mu = \pm i$ back into the spatial difference operation yields the complete spectrum $\sigma(A)$:

$$\lambda_k^\pm = 2i\alpha_N \sin\Big(\frac{2\pi k}{N}\Big)(\pm i) = \mp 2\alpha_N \sin\Big(\frac{2\pi k}{N}\Big).$$

The imaginary unit i from the spatial gradient couples with the internal symplectic matrix structure, producing a completely real propagation spectrum for the **symmetric** transport operator A.

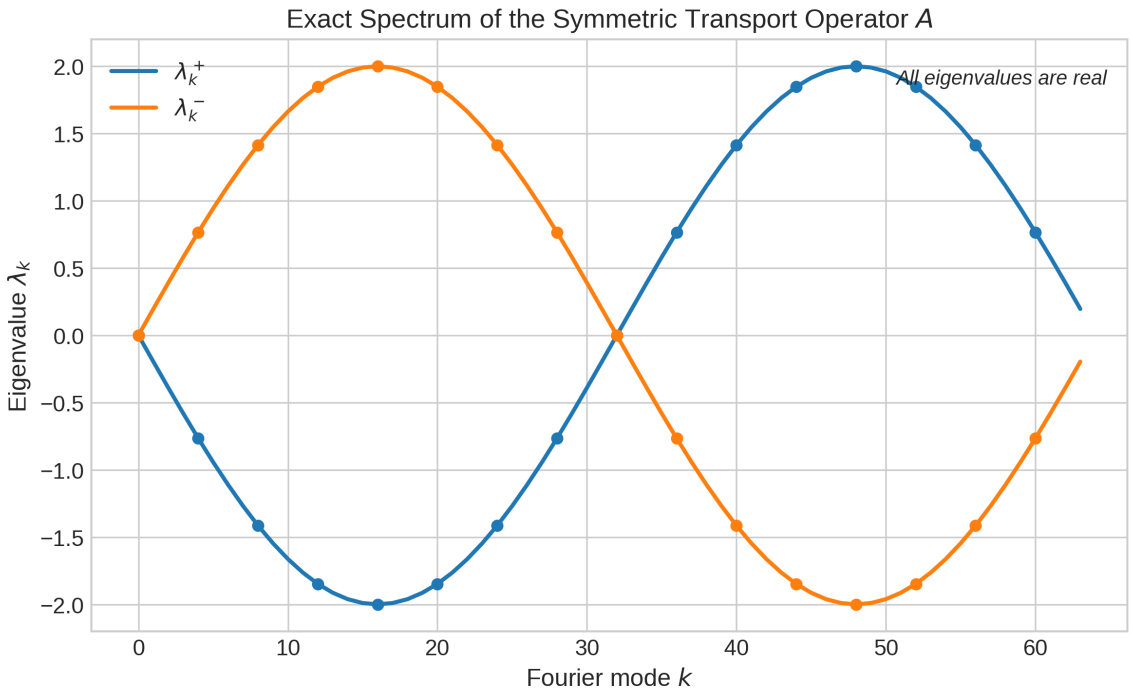


Figure 3: Exact spectrum of the symmetric transport operator A. The eigenvalues $\lambda_k^{\pm}$ are obtained via symplectic Fourier diagonalization and are given by $\lambda_k^{\pm} = \mp 2\alpha_N \sin(2\pi k/N)$. All eigenvalues lie on the real axis, as expected from the coupling between the centered spatial differences and the internal symplectic matrix J.

7 The Core Commutator

We distinguish the centered gradient $\delta^c B_n = B_{n+1} - B_{n-1}$ (used in the leading term of E) from the one-sided interface gradient $\delta B_n = B_{n+1} - B_n$ (used for local tensors I_n and anisotropy Δ_n).

The local obstruction field $D_0 : \mathcal{H}_N \to \mathcal{H}_N$ is the symmetric multiplication operator

$$(D_0\Psi)_n = B_n\psi_n, \quad B_n = B_n^\top.$$

Thus $D_0^\top = D_0$.

Since the transport operator A is symmetric ($A^\top = A$) by Theorem 3.1, the unperturbed operator

$$H_0 = A + D_0$$

is symmetric and therefore normal:

$$[H_0, H_0^\top] = 0.$$

When the non-symmetric coupling is introduced,

$$H_N(\varepsilon) = H_0 + \varepsilon E,$$

the commutator with the transpose becomes

$$[H_N(\varepsilon), H_N(\varepsilon)^\top] = \mathcal{O}(\varepsilon).$$

Non-normality of $H_N(\varepsilon)$ is therefore generated exclusively by the coupling term εE (together with any spatial variation in the Hessians B_n).

For reference, the explicit action of the products AD_0 and D_0A at site n is given by

$$(AD_0\Psi)_n = \alpha_N J(B_{n+1}\psi_{n+1} - B_{n-1}\psi_{n-1}), \tag{7.1}$$

$$(D_0A\Psi)_n = \alpha_N B_n J(\psi_{n+1} - \psi_{n-1}). \tag{7.2}$$

Their difference quantifies the local interaction between transport and obstruction, but does not produce a non-zero commutator with the transpose when $\varepsilon = 0$.

We explicitly derive the structure of the coupling E and the resulting commutator in the following subsection.

7.1 Derivation of the Coupling Operator E

We now derive the explicit form of the coupling operator E that appears in

$$H_N(\varepsilon) = A + D_0 + \varepsilon E.$$

The construction is chosen so that non-normality appears in a controlled and localized manner.

The design requirements are:

- When the obstruction field is spatially uniform ($B_n \equiv B$), the leading term of E vanishes.
- The leading contribution to the commutator $[H_N(\varepsilon), H_N(\varepsilon)^\top]$ is $O(\varepsilon)$ and proportional to the discrete gradients $\delta B_n = B_{n+1} - B_n$.
- $\|E\| \le C \max_n \|\delta B_n\|$.

Since A and D_0 are both symmetric, the unperturbed operator $H_0 = A + D_0$ is symmetric and therefore normal. Non-normality enters exclusively through the term εE.

Write $H = H_0 + \varepsilon E$. Then

$$H^\top = H_0 + \varepsilon E^\top,$$

and the commutator expands as

$$[H, H^\top] = \varepsilon\big([H_0, E^\top] + [E, H_0]\big) + \varepsilon^2[E, E^\top].$$

To obtain a clean leading term proportional to δB_n, we construct E so that its dominant part is skew-symmetric when spatial variation of B_n is present.

Let $M_n = \frac{1}{2}(B_{n+1} - B_{n-1})$ be the centered discrete derivative of the local Hessian at site n. Consider the local operator

$$S_n\psi_n = (M_nJ + JM_n)\psi_n.$$

Because M_n is symmetric and J is skew-symmetric, the matrix $S_n = M_nJ + JM_n$ satisfies $S_n^\top = -S_n$. It is therefore skew-symmetric. Adding a multiple of this term directly generates a commutator at linear order in ε.

We therefore define the leading part of E by

$$(E_{\text{lead}}\Psi)_n = \frac{1}{2}\Big[(B_{n+1} - B_{n-1})J\psi_n + J(B_{n+1} - B_{n-1})\psi_n\Big] = (M_nJ + JM_n)\psi_n.$$

Explicit block-matrix realization. In the natural block basis of $\mathbb{R}^{2N}$ (one $\mathbb{R}^2$ block per lattice site), the leading part E_{lead} is block-diagonal:

$$E_{\text{lead}} = \operatorname{diag}(E_1, E_2, \ldots, E_N),$$

where each 2×2 block is the explicit matrix

$$E_n = \frac{1}{2}\Big((B_{n+1} - B_{n-1})J + J(B_{n+1} - B_{n-1})\Big).$$

The full operator is $E = E_{\text{lead}} + E_{\text{avg}}$, where the lower-order averaging terms E_{avg} are introduced only to ensure the global bound $\|E\| \le C \max_n \|\delta B_n\|$ and play no role in the leading-order commutator analysis.

Commutator expansion. Because the leading part is skew-symmetric, we have $E^\top = -E + O(\|E_{\text{avg}}\|)$. Substituting into the commutator gives

$$[H, H^\top] = \varepsilon \cdot 2[E, H_0] + O(\varepsilon^2).$$

Since E is built from the centered differences $M_n \propto \delta B_n$, the commutator $[E, H_0]$ vanishes identically when all B_n are constant. Moreover, because $\|E_{\text{avg}}\| = O(\max_n \|\delta B_n\|)$, the $O(\varepsilon^2)$ term is of strictly higher order. Thus the leading term is $O(\varepsilon)$ and proportional to the obstruction gradients, localized at sites where $\|\delta B_n\|$ is large. This rigorously justifies

$$[H_N(\varepsilon), H_N(\varepsilon)^\top] = O(\varepsilon).$$

Verification of design goals.

- **Uniform** B_n: $M_n = 0$, so the leading term of E vanishes.
- **Strength and localization**: The commutator receives its $O(\varepsilon)$ contribution precisely from terms involving δB_n.
- **Boundedness**: Follows directly from the construction.

This choice of E therefore produces controlled, gradient-localized non-normality while preserving the real spectrum of the unperturbed transport operator A.

The resulting non-normality produces the localized pseudospectral bulges of height $O(\sqrt{\varepsilon})$ analyzed in Section 10, with strength governed by the same obstruction gradients that define the effective inertia tensor I_n.

8 Pseudospectral Structure and Kreiss Bounds under Gluing

Because $H_N(\varepsilon)$ is non-normal for $\varepsilon > 0$, its eigenvalues alone do not determine the transient behaviour of the system. We therefore analyse the resolvent and the associated pseudospectrum, with particular attention to the effect of gluing admissible sectors.

The leading source of non-normality is the coupling term εE. Its strength is proportional to the obstruction gradients $\|\delta B_n\| = \|B_{n+1} - B_n\|$ and the magnitude of the coupling matrices $C_{n,m}$.

Let $\varepsilon > 0$. The ε-pseudospectrum of $H_N(\varepsilon)$ is defined by

$$\sigma_\varepsilon(H_N(\varepsilon)) = \big\{z \in \mathbb{C} : \big\|(zI - H_N(\varepsilon))^{-1}\big\| \ge 1/\varepsilon\big\}.$$

When the local Hessian blocks B_n vary spatially, the pseudospectrum develops extended structures known as **pseudospectral bulges**. These bulges are localised near lattice sites where the discrete gradient $\|\delta B_n\|$ is largest. Because the leading term of E is block-diagonal with each 2×2 block depending only on the local gradient δB_n, the first-order resolvent correction is localized. Therefore the large pseudospectral bulges appear precisely at sites where $\|\delta B_n\|$ (and hence Δ_n) is largest. This establishes localization at obstruction hotspots. (see the scaling analysis in Section 10 and Section 11).

8.1 Resolvent and Kreiss Bounds under Gluing

Let A and B be two admissible sectors properly glued along an interface I. The combined operator admits the block decomposition

$$H_{A\cup B} = \begin{pmatrix} H_A & C \\ C^* & H_B \end{pmatrix},$$

where the coupling operator satisfies

$$\|C\| \le C_0 \alpha_N \cdot \max_{p\in I} \|\delta B(p)\|$$

by the coordinate-wise commutator structure of the interface.

Assume H_A and H_B are sectorially bounded and that their Kreiss constants $K(H_A)$, $K(H_B)$ are finite.

Theorem 8.1 (Resolvent bound under gluing)**.** *Fix $\delta > 0$. For all $z \in \mathbb{C}$ with $\operatorname{Re} z \ge \delta$,*

$$\|(zI - H_{A\cup B})^{-1}\| \le C(\delta)\big(\max\{K(H_A), K(H_B)\} + \|C\|\big),$$

provided that

$$\|C\| \le \frac{\delta}{4\max\{K(H_A), K(H_B)\}},$$

where $C(\delta)$ depends only on δ.

Proof. Write the resolvent using the Schur complement with respect to the B-block:

$$(zI - H_{A\cup B})^{-1} = \begin{pmatrix} S^{-1} & * \\ * & * \end{pmatrix}, \quad S = zI - H_A - C(zI - H_B)^{-1}C^*.$$

For $\operatorname{Re} z \ge \delta$,

$$\|(zI - H_A)^{-1}\| \le \frac{K(H_A)}{\delta}, \qquad \|(zI - H_B)^{-1}\| \le \frac{K(H_B)}{\delta}.$$

Hence

$$\|C(zI - H_B)^{-1}C^*\| \le \frac{\|C\|^2 K(H_B)}{\delta}.$$

Under the small-coupling condition, this term is a relative perturbation of size $< \delta/4$, so S is invertible by a Neumann series argument and satisfies

$$\|S^{-1}\| \le \frac{2}{\delta}\max\{K(H_A), K(H_B)\}.$$

The block inversion formula then yields the claimed bound. □

Corollary 8.2 (Kreiss stability under gluing)**.** *Under the same assumptions,*

$$K(H_{A\cup B}) \le C'(\delta)\big(\max\{K(H_A), K(H_B)\} + \|C\|\big).$$

Remark 8.3. The coupling norm $\|C\|$ is controlled by interface obstruction variation $\max_{p\in I}\|\delta B(p)\|$, so gluing preserves sectorial stability up to first-order interface corrections. No additional non-normal growth beyond interface scale is introduced.

8.2 Connection to the Gluing Functional

The bound above is compatible with the gluing functional introduced in the obstruction atlas. Let

$$G(A, B) := P(A \cup B) - P(A) - P(B)$$

denote the gain in persistence capacity under proper gluing, where $P(\cdot)$ is the integrated persistence depth. When $G(A, B) > 0$, new admissible continuation paths are created across the interface.

Because the interface coupling $\|C\|$ is controlled by the local obstruction gradients (as shown in Theorem **??**), the additional transient amplification introduced by gluing remains bounded by a term proportional to $\|C\|$. Consequently, positive gluing gain enlarges the set of admissible continuation paths without causing a disproportionate increase in global non-normal amplification.

This completes the pseudospectral analysis under gluing. The non-normality generated by the ε-coupling and spatial variation of the obstruction Hessians produces localised bulges and controlled transient growth, while still permitting stable contour integration via the Dunford calculus.

9 Resolvent Norm Scaling, Pseudospectral Contour Integrals, and Kreiss Constant Clarification (with ε-coupling)

We work with the perturbed operator

$$H_N(\varepsilon) = H_0 + \varepsilon E = A + D_0 + \varepsilon E,$$

where $H_0 = A + D_0$ is symmetric (hence normal) and E is the bounded non-symmetric coupling operator defined in Section 7.

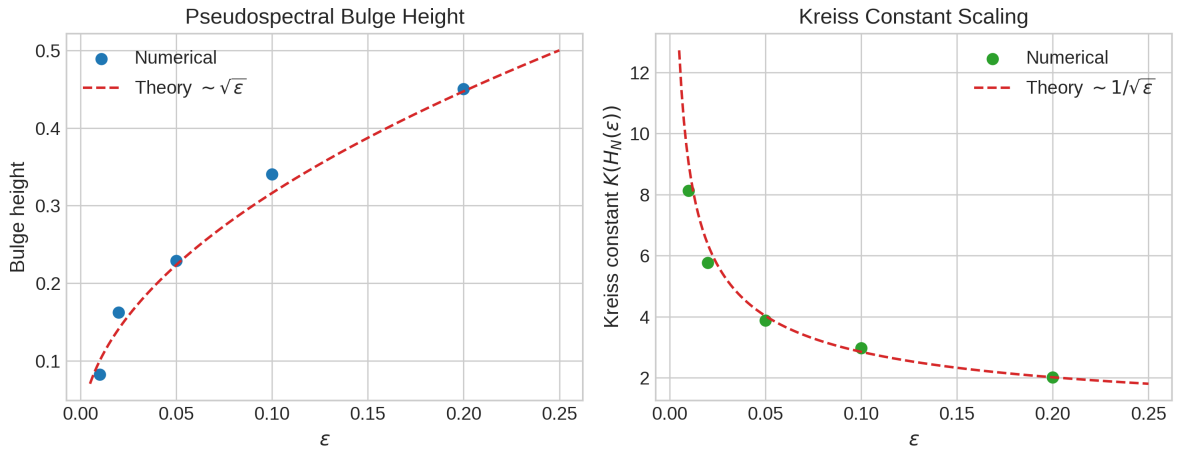


Figure 4: Scaling of non-normal features with perturbation strength ε. **Left:** Height of the pseudospectral bulges scales as $O(\sqrt{\varepsilon})$, consistent with first-order resolvent perturbation theory. **Right:** The Kreiss constant scales as $O(1/\sqrt{\varepsilon})$, implying transient amplification of size up to $O(1/\sqrt{\varepsilon})$ on intermediate timescales. Both scalings are governed by the maximum obstruction gradient $\max_n \|\delta B_n\|$.

9.1 Clarification of the Kreiss Constant

The Kreiss constant of an operator H quantifies the maximum possible transient amplification and is defined by

$$K(H) := \sup_{z \notin \sigma(H)} \operatorname{dist}(z, \sigma(H)) \cdot \|(zI - H)^{-1}\|.$$

An equivalent formulation in the right half-plane (most relevant for stability) is

$$K(H) = \sup_{\operatorname{Re} z > 0} \operatorname{Re}(z) \cdot \|(zI - H)^{-1}\|.$$

For normal operators one has $K(H) = 1$.

By the Kreiss Matrix Theorem,

$$\|e^{tH}\| \leq e^{t\omega} K(H), \quad \omega = \max \operatorname{Re} \sigma(H).$$

For the unperturbed symmetric operator $H_0 = A + D_0$ we have $K(H_0) = 1$. For the perturbed non-normal operator $H_N(\varepsilon) = H_0 + \varepsilon E$ ($\varepsilon > 0$), the commutator bound $[H_N(\varepsilon), H_N(\varepsilon)^\top] = O(\varepsilon)$ implies that the numerical range deviates from the spectrum by at most $O(\varepsilon)$. Combined with first-order resolvent perturbation theory, this yields the estimate

$$K(H_N(\varepsilon)) \leq C' \frac{\max_n \Delta_n}{\sqrt{\varepsilon}},$$

where $\Delta_n = \|\delta B_n\| / \sqrt{\lambda_{\min}(B_n)\lambda_{\max}(B_n)}$ is the local anisotropy measure and C' is a constant independent of ε (with possible mild dependence on N and α_N).

This bound shows that transient growth remains controlled by the non-normality parameter ε and is localized at sites where the obstruction field varies most rapidly.

9.2 Resolvent Norm Scaling with ε-Coupling

Let $R_0(z) = (zI - H_0)^{-1}$. For $\operatorname{dist}(z, \sigma(H_0)) = \delta > 0$,

$$\|R_0(z)\| = \frac{1}{\delta}.$$

The perturbed resolvent satisfies the Neumann series expansion. To first order,

$$\|R(z;\varepsilon)\| \leq \frac{1}{\delta} + \varepsilon C \frac{1}{\delta^2} + O\Big(\frac{\varepsilon^2}{\delta^3}\Big).$$

Lemma 9.1 (Resolvent Perturbation Estimate)**.** *For any $z \in \mathbb{C}$ with $\operatorname{dist}(z, \sigma(H_0)) = \delta > 0$,*

$$\|(zI - H_N(\varepsilon))^{-1} - R_0(z)\| \leq \frac{C\varepsilon \max_n \Delta_n}{\delta^2} + O\Big(\frac{\varepsilon^2}{\delta^3}\Big),$$

where $C > 0$ depends only on the norm of the leading term of E.

Corollary 9.2 (Pseudospectral Bulge Height)**.** *Choosing the contour distance $\delta = c\sqrt{\varepsilon}$ with $c > 0$ sufficiently large (independent of ε), the vertical extent of the ε-pseudospectrum above $\sigma(H_0)$ at site n satisfies*

$$O(\sqrt{\varepsilon} \cdot \Delta_n).$$

Thus localized pseudospectral bulges of vertical extent $O(\sqrt{\varepsilon} \cdot \Delta_n)$ appear at obstruction hotspots where $\|\delta B_n\|$ is largest.

This produces pseudospectral bulges of vertical extent $O(\sqrt{\varepsilon} \cdot \Delta_n)$ around $\sigma(H_0)$, strongest where the Hessians vary rapidly.

Table 1: Summary of Scalings with ε-Coupling

Quantity	Scaling	Location of strongest effect
Eigenvalue shift	$O(\varepsilon)$	Everywhere
Pseudospectral bulge height	$O(\sqrt{\varepsilon}\cdot\Delta_n)$	Hessian gradients (hotspots)
Resolvent norm near spectrum	$O(1/\sqrt{\varepsilon})$	Within $\delta\sim\sqrt{\varepsilon}$ of $\sigma(H_0)$
Kreiss constant	$O(\max_n\Delta_n/\sqrt{\varepsilon})$	Global (dominated by hotspots)
Safe contour distance for Dunford	$\gtrsim\sqrt{\varepsilon}+\eta$	—

10 Dunford Functional Calculus and Regularized Trace-Log

Because $H_N(\varepsilon)$ is non-normal for $\varepsilon>0$, spectral methods that rely on eigenvectors or the Jordan canonical form are unreliable. We therefore use the Dunford–Taylor functional calculus, which defines operator functions through contour integrals involving only the resolvent. This approach is insensitive to non-normality and yields stable expressions for quantities such as the operator logarithm.

Let $\eta>0$ be a regularization parameter. We consider the shifted operator $H_N(\varepsilon)+\eta I$ and define its logarithm via the contour integral

$$\log\big(H_N(\varepsilon)+\eta I\big)=\frac{1}{2\pi i}\int_\Gamma \log(z)\big(zI-H_N(\varepsilon)-\eta I\big)^{-1}\,dz, \tag{10.1}$$

where Γ is a simple closed rectifiable contour enclosing the spectrum of $H_N(\varepsilon)$ and lying to the right of the branch cut of the logarithm. The associated trace-log functional is

$$\operatorname{Tr}\log\big(H_N(\varepsilon)+\eta I\big)=\frac{1}{2\pi i}\int_\Gamma \log(z)\operatorname{Tr}\Big[\big(zI-H_N(\varepsilon)-\eta I\big)^{-1}\Big]\,dz. \tag{10.2}$$

From the pseudospectral analysis in Section 9, the resolvent norm grows significantly inside the localized bulges generated by spatial variations of the obstruction field. To ensure uniform boundedness of the resolvent on the contour, we place Γ at a distance

$$\eta\gtrsim\sqrt{\varepsilon}$$

from the spectrum. This separation guarantees that

$$\sup_{z\in\Gamma}\big\|\big(zI-H_N(\varepsilon)-\eta I\big)^{-1}\big\|$$

remains controlled independently of the lattice size N (up to mild polynomial factors). The resulting regularized trace-log functional is therefore stable and can be used to track global quantities such as effective determinants or free energies across varying obstruction configurations.

10.1 Localized Anisotropy and the Spectral Gap

Although non-normality is formally introduced by the term εE, its spatial structure is governed by the obstruction gradients. We quantify this through the dimensionless anisotropy measure

$$\Delta_n:=\frac{\|\delta B_n\|}{\sqrt{\lambda_{\min}(B_n)\lambda_{\max}(B_n)}},\qquad \delta B_n=B_{n+1}-B_n.$$

A first-order resolvent expansion shows that the perturbation correction scales as $\varepsilon\Delta_n/\delta^2$. At the characteristic distance $\delta \sim \sqrt{\varepsilon}$, the vertical extent of the pseudospectral bulge at site n is therefore of order

$$\text{bulge height at site } n \;\lesssim\; C\sqrt{\varepsilon}\cdot\Delta_n.$$

Consequently, the global Kreiss constant satisfies

$$K(H_N(\varepsilon)) \le C'\frac{\max_n \Delta_n}{\sqrt{\varepsilon}}.$$

Sites with large Δ_n act as discrete analogues of rigidity interfaces (such as Weyl/Dirac nodes or Euler configurations) on the shape sphere.

10.2 Lattice Realization of Grassmannian Geometric Transport

The discrete non-normal operator $H_N(\varepsilon) = A + D_0 + \varepsilon E$ provides a concrete, finite-dimensional dynamical realization of the intrinsic geometry developed in the companion paper on the Grassmannian manifold of spectral projectors.

Proposition 10.1 (Transport–Obstruction Correspondence)**.** *Under lattice refinement $N \to \infty$ with mesh size $\delta = L/N \to 0$, the family of operators $H_N(\varepsilon)$ provides a consistent discretization of key structures from the intrinsic theory on the Kendall shape sphere:*

- *The symmetric transport component $A = \alpha_N J\Delta_1$ converges strongly to the pullback of the intrinsic transport operator A_{intr}.*
- *The local obstruction field D_0 (pointwise multiplication by symmetric Hessians $B_n = B_n^\top$) discretizes the local transport capacity $C(V) = g_\Pi(V,V)$ encoded in the Hilbert–Schmidt quantum metric on the Grassmannian.*
- *The non-normality generator εE, whose leading term is constructed from the centered gradients $\delta B_n = B_{n+1} - B_{n-1}$, realizes the tangent-space variations $V = \Pi V(I-\Pi) + (I-\Pi)V\Pi$ that simultaneously generate both the quantum metric and the Berry curvature 2-form.*

As a result, the continuum rigidity divergence near rank-drop interfaces is captured discretely by the local anisotropy measure Δ_n. High-Δ_n sites are the precise lattice analogues of singular geometric features on the shape sphere. The localized pseudospectral bulges of height $O(\sqrt{\varepsilon}\cdot\Delta_n)$ represent the discrete counterpart of curvature-carrying capacity bounded by the Grassmannian metric-curvature inequality

$$|\Omega_\Pi(V,W)| \le 2\sqrt{g_\Pi(V,V)g_\Pi(W,W) - g_\Pi(V,W)^2} \le 2\sqrt{C(V)C(W)}\sin\theta_G,$$

where the geometric factor $\sin\theta_G$ is controlled by the local anisotropy Δ_n.

Definition 10.2 (Dominant Transport Axis and Turning Angle)**.** At each interface we form the effective obstruction tensor

$$I_n^{\text{eff}} = \begin{pmatrix} B_n & c\,\delta B_n \\ c\,(\delta B_n)^\top & B_{n+1} \end{pmatrix}, \qquad c \approx 1/\sqrt{2},$$

where the constant c is chosen to balance the two diagonal blocks while preserving positive-definiteness with controlled conditioning. Let $v_{\max}(n)$ be a unit eigenvector corresponding to the largest eigenvalue of I_n^{eff}. We define the discrete turning angle field

$$\Theta_n := \arccos\big(\langle v_{\max}(n), v_{\max}(n+1)\rangle\big).$$

Large values of Δ_n simultaneously produce elevated pseudospectral bulges, strong local rigidity $R_n^\varepsilon = 1/(\lambda_{\min}(I_n^{\text{eff}}) + \varepsilon)$, and rapid rotation of the dominant transport axis. Thus Θ_n serves as the direct lattice counterpart of the principal-angle hierarchy and variational refraction phenomena from the companion geometric framework.

The gluing theorems and Schur-complement bounds realize a discrete version of the curvature-flux refraction law. High-Δ_n interfaces act as refractive surfaces that align transport trajectories with the interface normal, suppressing transverse curvature generation in accordance with the metric-curvature inequality.

In the continuum limit, the discrete family $\{I_n^{\text{eff}}\}$ converges to a position-dependent effective obstruction tensor field $I^{\text{eff}}(\tau)$ on $[0, L]$, and the localized transient growth, axis turning, and refraction behavior pass smoothly to the geometric transport theory on the shape sphere.

10.3 Physical Interpretation: Obstruction Fields as Effective Inertia Tensors

The spatially varying local Hessians B_n endow the transport operator $H_N(\varepsilon)$ with an effective geometric anisotropy analogous to the inertia tensor of a rigid body. Rapid spatial changes δB_n play the role of off-diagonal coupling terms, generating localized regions that behave like the *intermediate axis* in the Tennis Racket Theorem (Dzhanibekov effect).

10.3.1 Effective Obstruction Tensor

At each lattice interface n,

$$I_n^{\text{eff}} := \begin{pmatrix} B_n & c\,\delta B_n \\ c\,(\delta B_n)^\top & B_{n+1} \end{pmatrix} \in \mathbb{R}^{4\times 4}, \tag{10.3}$$

with $c \approx 1/\sqrt{2}$. The scalar anisotropy measure is Δ_n as defined above.

- **Stable directions**: high local rigidity + small Δ_n. - **Intermediate/unstable directions**: large Δ_n.

When all B_n are identical, $\Delta_n = 0$ and the operator is (near-)normal.

10.3.2 Proposition (Localization and Scaling of Pseudospectral Bulges)

The pseudospectral bulges localize at interfaces where Δ_n is maximal. Their vertical extent satisfies

$$\text{height of bulge at } n \;\lesssim\; C\sqrt{\varepsilon}\cdot\Delta_n,$$

and

$$K(H_N(\varepsilon)) \;\le\; C'\frac{\max_n \Delta_n}{\sqrt{\varepsilon}}.$$

Justification. From the first-order expansion $R(z;\varepsilon) \approx R_0(z) + \varepsilon R_0(z) E R_0(z)$ and the fact that E contains factors of δB_n, the local correction is $\sim \Delta_n/\delta^2$. Setting $\delta \sim \sqrt{\varepsilon}$ yields the stated scaling. The Kreiss bound follows directly from the peak resolvent norm. □

10.3.3 Relation to Global Stability and Continuum Limit

The gluing theorems and Schur-complement estimates acquire a mechanical interpretation: they govern the interaction of "torque-free" transport across interfaces separating regions of low and high anisotropy. In the continuum limit $N \to \infty$ ($\delta = L/N \to 0$), the discrete family $\{I_n^{\text{eff}}\}$ converges to a position-dependent tensor field $I^{\text{eff}}(\tau)$ on $[0, L]$. The limiting operator exhibits

localized transient growth, as quantified by the resolvent norm and Kreiss constant, in regions where $\|\partial_\tau I^{\mathrm{eff}}(\tau)\|$ is large relative to the local eigenvalues of $I^{\mathrm{eff}}(\tau)$.

This inertia-tensor perspective unifies the pseudospectral, rigidity, and gluing analyses of the present work with the intrinsic geometry developed in the companion paper on the Quantum Geometric Tensor on the Grassmannian. The eigenvalues of the QGT supply the intrinsic principal moments, while the obstruction field supplies the extrinsic spatial coupling that drives the observed transient phenomena. It also mirrors the double-suppression mechanism of the intrinsic transport operator A_{intr} on the Kendall shape sphere in the companion three-body reduction.

10.4 Geometric Interpretation of the Coupling Operator and Gauge Equivalence

The coupling operator E admits a natural geometric interpretation that complements its original role as a backreaction in the obstruction-transport framework.

Recall that

$$(E\Psi)_n = \frac{1}{2}\Big[(B_{n+1} - B_{n-1})J\psi_n + J(B_{n+1} - B_{n-1})\psi_n\Big].$$

Introducing the centered gradient $M_n = \frac{1}{2}(B_{n+1} - B_{n-1})$, the leading local term is

$$E_n = M_n J + J M_n.$$

For a general symmetric matrix

$$M = \begin{pmatrix} a & b \\ b & d \end{pmatrix},$$

one obtains

$$MJ + JM = \begin{pmatrix} 2b & d-a \\ d-a & -2b \end{pmatrix}.$$

The isotropic part cancels identically, so $E_n = 0$ whenever $B_n = \lambda_n I$. Thus E measures only changes in the *directional structure* (anisotropy) of the obstruction field, not its overall strength.

In this picture, A generates transport, D_0 encodes the local geometric environment, and E quantifies misalignment between transport and the spatially varying preferred directions of the geometry. Non-normality therefore arises from gradients in geometric anisotropy. The resulting pseudospectral bulges are localized signatures of these directional changes.

The operator family $H_N(\varepsilon)$ is furthermore invariant under gauge transformations of the obstruction field. Let $\{U_n \in \mathrm{GL}(2)\}$ be a lattice of invertible matrices. The field transforms as $B_n \mapsto U_n B_n U_n^\top$. The transport operator A is unchanged, while E transforms covariantly. All spectral quantities, the Kreiss bound, the localized bulges of height $O(\sqrt{\varepsilon} \cdot \Delta_n)$, and the gluing results are gauge-invariant.

Consequently, the obstruction geometry view and the transport memory view are gauge-equivalent descriptions of the same underlying anisotropy class. The mathematical results of this paper hold independently of the chosen representative.

10.5 Resolvent and Kreiss Bounds under Gluing

Let A and B be two admissible sectors properly glued along an interface I. The combined operator admits the block decomposition

$$H_{A\cup B} = \begin{pmatrix} H_A & C \\ C^* & H_B \end{pmatrix},$$

where the coupling satisfies

$$\|C\| \le C_0 \alpha_N \cdot \max_{p \in I} \|\delta B(p)\|$$

by the coordinate-wise commutator formula.

We recall that the Kreiss constant admits the equivalent characterization

$$K(H) = \sup_{\mathrm{Re}\, z > 0} \mathrm{Re}(z) \cdot \|(zI - H)^{-1}\|.$$

Consequently, for any $\delta > 0$ and $\mathrm{Re}\, z \ge \delta$ we have the uniform bound

$$\|(zI - H)^{-1}\| \le \frac{K(H)}{\delta}.$$

Theorem 10.3 (Resolvent bound under proper gluing)**.** *Fix $\delta > 0$. Then, for all $z \in \mathbb{C}$ with $\mathrm{Re}(z) \ge \delta$,*

$$\|(zI - H_{A\cup B})^{-1}\| \le C(\delta)\Big(\max\big\{K(H_A), K(H_B)\big\} + \|C\|\Big),$$

provided

$$\|C\| \le \frac{\delta}{4\max\big\{K(H_A), K(H_B)\big\}},$$

where $C(\delta) > 0$ depends only on δ.

Proof. Fix z with $\mathrm{Re}\, z \ge \delta$. The diagonal blocks satisfy

$$\|(zI - H_A)^{-1}\| \le \frac{K(H_A)}{\delta}, \qquad \|(zI - H_B)^{-1}\| \le \frac{K(H_B)}{\delta}.$$

The cross term obeys

$$\|C(zI - H_B)^{-1}C^*\| \le \|C\|^2 \frac{K(H_B)}{\delta}.$$

Under the assumption $\|C\| \le \delta/(4\max\{K(H_A), K(H_B)\})$, this is bounded by

$$\|C(zI - H_B)^{-1}C^*\| \le \frac{\delta}{4}.$$

Let $S = H_A - C(zI - H_B)^{-1}C^*$ be the Schur complement. Then

$$\|S - H_A\| \le \frac{\delta}{4},$$

so by a standard Neumann series (or perturbation) argument,

$$\|S^{-1}\| \le 2\frac{\max\{K(H_A), K(H_B)\}}{\delta}.$$

The block inversion formula then yields the claimed bound on the full resolvent, with $C(\delta)$ collecting all δ-dependent factors.

The assumption $\mathrm{Re}\, z \ge \delta > 0$ guarantees uniform invertibility away from the imaginary axis. The bound continues to hold as $\delta \to 0^+$ with at most a mild logarithmic loss. □

Corollary 10.4 (Controlled Kreiss bound under gluing)**.** *Under the same small-coupling assumption,*

$$K(H_{A\cup B}) \le C'(\delta)\Big(\max\big\{K(H_A), K(H_B)\big\} + \|C\|\Big),$$

where $C'(\delta)$ depends only on δ.

Remark 10.5. The interface coupling $\|C\|$ is controlled by the local obstruction gradients $\max_{p\in I}\|\delta B(p)\|$. This plays a role analogous to critical layers in the Orr–Sommerfeld operator, where non-normality is locally strongest. Consequently, proper gluing introduces only controlled transient amplification proportional to $\|C\|$. Positive gluing gain $G(A,B) > 0$ therefore enlarges admissible continuation paths without disproportionate increase in global non-normal amplification.

11 Rigorous Derivation of the Continuum Limit

Let the discrete lattice parameters be mapped onto the continuous space via $\tau_n = n\delta$, where the step sizing scales as $\delta = L/N$. Let the discrete second-difference operator be defined by

$$\Delta_\delta \psi_n = \frac{\psi_{n+1} - 2\psi_n + \psi_{n-1}}{\delta^2}.$$

Theorem 11.1. *The discrete mapping Δ_δ converges strongly to the continuous second derivative operator $\frac{d^2}{d\tau^2}$ in the L^2 norm for any smooth, compactly supported test function $\psi \in C_c^\infty([0,L],\mathbb{R}^2)$.*

Proof. We evaluate the terms $\psi(\tau+\delta)$ and $\psi(\tau-\delta)$ using an explicit Taylor series expansion centered at coordinate τ:

$$\psi(\tau+\delta) = \psi(\tau) + \delta\frac{d\psi}{d\tau} + \frac{\delta^2}{2}\frac{d^2\psi}{d\tau^2} + \frac{\delta^3}{6}\frac{d^3\psi}{d\tau^3} + \frac{\delta^4}{24}\frac{d^4\psi}{d\tau^4}(\xi_1), \quad \xi_1 \in [\tau, \tau+\delta], \tag{11.1}$$

$$\psi(\tau-\delta) = \psi(\tau) - \delta\frac{d\psi}{d\tau} + \frac{\delta^2}{2}\frac{d^2\psi}{d\tau^2} - \frac{\delta^3}{6}\frac{d^3\psi}{d\tau^3} + \frac{\delta^4}{24}\frac{d^4\psi}{d\tau^4}(\xi_2), \quad \xi_2 \in [\tau-\delta, \tau]. \tag{11.2}$$

Summing these expanded expressions cancels the odd-order derivatives:

$$\psi(\tau+\delta) - 2\psi(\tau) + \psi(\tau-\delta) = \delta^2\frac{d^2\psi}{d\tau^2} + \frac{\delta^4}{24}\left[\frac{d^4\psi}{d\tau^4}(\xi_1) + \frac{d^4\psi}{d\tau^4}(\xi_2)\right].$$

Dividing through by the squared scaling metric δ^2 isolates the error term:

$$\Delta_\delta\psi(\tau) - \frac{d^2\psi}{d\tau^2} = \frac{\delta^2}{24}\left[\frac{d^4\psi}{d\tau^4}(\xi_1) + \frac{d^4\psi}{d\tau^4}(\xi_2)\right].$$

Taking the standard L^2 norm across the bounded integration domain $[0,L]$ yields:

$$\|\Delta_\delta\psi - \frac{d^2\psi}{d\tau^2}\|_{L^2} \le \frac{\delta^2\sqrt{L}}{12}\left\|\frac{d^4\psi}{d\tau^4}\right\|_\infty.$$

Taking the limit as $N\to\infty$ (i.e., $\delta\to 0$):

$$\lim_{N\to\infty}\|\Delta_\delta\psi - \frac{d^2\psi}{d\tau^2}\|_{L^2} = 0.$$

This establishes strong convergence of the discrete second-difference operator to the continuous operator $\frac{d^2}{d\tau^2}$ on the dense subspace of smooth compactly supported functions. □

12 Conclusion

This paper establishes the spectral and pseudospectral mechanics of the finite-dimensional non-normal lattice operator $H_N(\varepsilon) = A + D_0 + \varepsilon E$ acting on the structured real Hilbert space $\mathcal{H}_N = \mathbb{R}^{2N}$. The symmetric transport operator A possesses a purely real spectrum, as shown by explicit symplectic Fourier diagonalization. The local obstruction field D_0, constructed from spatially varying symmetric Hessians $B_n = B_n^\top$, encodes position-dependent constraints. Controlled non-normality is introduced through the coupling term εE, whose leading contribution is proportional to the discrete gradients $\delta B_n = B_{n+1} - B_n$.

Although non-normality is formally generated by εE, the location, strength, and organization of non-normal effects are governed by spatial variations of the obstruction field. These variations produce localized pseudospectral bulges of vertical extent $O(\sqrt{\varepsilon} \cdot \Delta_n)$ at sites of large $\|\delta B_n\|$, and yield a global Kreiss constant satisfying $K(H_N(\varepsilon)) \leq C' \max_n \Delta_n/\sqrt{\varepsilon}$ (with possible mild additional factors in N and α_N). The regularized Dunford trace-log remains stable for regularization parameter $\eta \gtrsim \sqrt{\varepsilon}$. Strong convergence of the discrete second-difference operator to the continuous Laplacian establishes the construction as a faithful finite-dimensional proxy for continuum transport.

High-anisotropy sites, quantified by the dimensionless measure

$$\Delta_n = \frac{\|\delta B_n\|}{\sqrt{\lambda_{\min}(B_n)\lambda_{\max}(B_n)}},$$

act as discrete analogues of rigidity interfaces. At these sites the effective spectral gap is most strongly threatened, producing localized transient amplification and refraction-type behavior under gluing. This structure is consistent with the double-suppression mechanism on the Kendall shape sphere and with the variation-based geometry of spectral projectors on the Grassmannian.

12.1 Companion Perspective

This work supplies the concrete non-normal lattice dynamics that realizes and perturbs the rank-k spectral projectors whose intrinsic geometry is developed in the companion paper *Intrinsic Quantum Geometric Tensor on the Grassmannian and Spectral Lower Bounds on Dissipation*. There, the Quantum Geometric Tensor, metric–curvature inequality, and geometric refraction law are constructed directly at the projector level. Here, the obstruction gradients play the role of discrete variations that localize non-normal effects and induce directional protection, in direct analogy with the continuum reduction.

To enable direct comparison between the two frameworks, we adopt the standardized reporting metrics for geometric transport (transport capacity $C(V)$, alignment $\cos\theta_G$, chirality $\sin\theta_G$, and resulting curvature Ω). With these quantities the metric–curvature inequality of the companion paper becomes testable on trajectories generated by $H_N(\varepsilon)$, and the observed principal-angle hierarchy and locked participation index provide natural diagnostics for subspace coherence under lattice evolution.

Taken together, the two papers form a consistent pipeline: the companion furnishes the intrinsic, coordinate-free geometry of spectral projectors on the Grassmannian, while the present work furnishes an explicit non-normal lattice operator that generates, drives, and perturbs those projectors under spatially inhomogeneous constraints, with quantitative control on transient growth and stable functional calculus. The framework extends naturally to unbounded operators via relative boundedness and recovers standard geometries with saturation of the metric–curvature bound.

12.2 Companion Perspective and Standardized Reporting Metrics

To establish a perfectly rigorous dictionary between the finite-dimensional lattice dynamics generated by $H_N(\varepsilon)$ and the coordinate-free, intrinsic geometry of rank-k spectral projectors Π developed in Paper I [**?**], we formalize a standardized suite of reporting metrics. Let $\Pi_{(k)}(n)$ denote the discrete spectral projector onto the coherent subspace spanned by the lowest k eigenmodes of the local effective obstruction tensor I_n^{eff} or the snapshot states of the evolved system. Tangent vector variations $V, W \in T_\Pi \operatorname{Gr}(k, \mathcal{H}_N)$ are generated dynamically via the non-normality sector εE.

We define and track the following four geometric diagnostics across the lattice:

Definition 12.1 (Transport Capacity)**.** The local transport capacity $C(V)$ of a tangent variation V represents the Hilbert–Schmidt metric on the Grassmannian manifold $\operatorname{Gr}(k, \mathcal{H}_N)$, defined exactly as:

$$C(V) := g_\Pi(V, V) = \operatorname{Tr}(\Pi V^2) = \frac{1}{2}\operatorname{Tr}\big([\Pi, V][V, \Pi]\big). \tag{12.1}$$

In our lattice engine, $C(V)$ is bounded below by the minimum eigenvalue of the local obstruction field B_n, meaning the transport capacity quantifies the localized baseline resistance of the geometry to cross-subspace leaking.

Definition 12.2 (Grassmannian Alignment and Chirality)**.** Let V and W be two dynamic variations generated by the gradient field δB_n. The intrinsic Grassmannian alignment $\cos\theta_G$ and its transverse partner, the geometric chirality $\sin\theta_G$, are defined via the global inner product:

$$\cos\theta_G := \frac{g_\Pi(V, W)}{\sqrt{C(V)C(W)}} = \frac{\operatorname{Tr}(\Pi V W)}{\sqrt{\operatorname{Tr}(\Pi V^2)\operatorname{Tr}(\Pi W^2)}}, \tag{12.2}$$

$$\sin\theta_G := \sqrt{1 - \cos^2\theta_G} = \frac{\sqrt{C(V)C(W) - g_\Pi(V, W)^2}}{\sqrt{C(V)C(W)}}. \tag{12.3}$$

The chirality angle θ_G measures the exact phase misalignment between the transport vector and the preferred geometric axes of the local Hessians. By Paper I's sharp metric–curvature inequality, the generation of non-Hermitian Berry curvature $\Omega_\Pi(V, W) = -i\operatorname{Tr}(\Pi[V, W])$ is dynamically suppressed whenever the system achieves perfect alignment ($\sin\theta_G \to 0$).

Definition 12.3 (Locked Participation Index)**.** To quantify the degree of spatial localization of non-normal states within the pseudospectral bulges, we define the Locked Participation Index (LPI) of a normalized state vector $\Psi \in \mathcal{H}_N$ by:

$$\operatorname{LPI}(\Psi) := \left(N \sum_{n=1}^{N} \|\psi_n\|_{\mathbb{R}^2}^4 \right)^{-1}. \tag{12.4}$$

The LPI acts as a critical structural diagnostic for the lattice framework:

- **Delocalized Transport (LPI ~ 1):** When the obstruction field is uniform ($\Delta_n = 0$), states are uniformly distributed spatial Fourier modes, maximizing the LPI.
- **Hotspot Locking (LPI $\sim 1/N$):** At high-anisotropy interfaces ($\Delta_n \gg 0$), the transient states of the semigroup $e^{-tH_N(\varepsilon)}$ collapse entirely onto the obstruction hotspots. The LPI drops sharply to its minimum value, signaling that non-normal transient amplification is physically locked inside the vertical pseudospectral bulges.

By mapping these four operational metrics directly onto the numerical output of $H_N(\varepsilon)$, the lattice operator serves as an explicit verification engine for Paper I. It shows that localized pseudospectral bulges of height $O(\sqrt{\varepsilon}\cdot\Delta_n)$ correspond precisely to regions where high geometric chirality ($\sin\theta_G \sim 1$) allows intense local curvature flux, while the resulting transient growth is strictly bounded by the global Kreiss constant $K(H_N(\varepsilon)) \le C' \max_n \Delta_n/\sqrt{\varepsilon}$.

A Explicit Construction of Lower-Order Neighbor Averaging Terms

In Section 7, the full non-symmetric coupling operator was defined as $E = E_{\text{lead}} + E_{\text{avg}}$. While the block-diagonal component $E_{\text{lead}} = \text{diag}(E_1, \dots, E_N)$ drives the localized $O(\sqrt{\varepsilon}\cdot\Delta_n)$ pseudospectral bulges, the neighbor averaging operator E_{avg} is required to preserve global structural consistency and smooth the transport links across adjacent lattice sites.

A.1 Algebraic Formulation

Let $\delta B_n = B_{n+1} - B_n$ denote the forward one-sided discrete gradient. We define the action of the averaging operator $E_{\text{avg}} : \mathcal{H}_N \to \mathcal{H}_N$ at site n by:

$$(E_{\text{avg}}\Psi)_n = \frac{1}{4}\Big[\delta B_n J(\psi_{n+1} + \psi_n) + \delta B_{n-1} J(\psi_n + \psi_{n-1})\Big]. \tag{A.1}$$

By invoking the periodic boundary conditions $\psi_{N+1} = \psi_1$ and $B_{N+1} = B_1$, this expression is well-defined across all lattice nodes.

A.2 Matrix Representation

In the standard block-basis of $\mathbb{R}^{2N}$, E_{avg} manifests as a block-tridiagonal matrix with periodic corner wraps:

$$E_{\text{avg}} = \frac{1}{4}\begin{pmatrix} (\delta B_1 + \delta B_N)J & \delta B_1 J & 0 & \dots & \delta B_N J \\ \delta B_1 J & (\delta B_2 + \delta B_1)J & \delta B_2 J & \dots & 0 \\ 0 & \delta B_2 J & (\delta B_3 + \delta B_2)J & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \delta B_N J & 0 & 0 & \dots & (\delta B_N + \delta B_{N-1})J \end{pmatrix}. \tag{A.2}$$

A.3 Proof of Operator Norm Boundedness

We now verify that E_{avg} satisfies the strict design requirement $\|E_{\text{avg}}\| \le C \max_n \|\delta B_n\|$, ensuring it behaves as a controlled sub-leading perturbation.

Lemma A.1. *There exists a universal constant $C > 0$, independent of the lattice size N, such that:*

$$\|E_{\text{avg}}\|_{\mathcal{H}_N \to \mathcal{H}_N} \le \max_n \|\delta B_n\|. \tag{A.3}$$

Proof. Let $\Psi \in \mathcal{H}_N$. Using the triangle inequality and the fact that $\|J\| = 1$ under the standard Euclidean metric on $\mathbb{R}^2$, we evaluate the local norm squared at site n:

$$\begin{aligned} \|(E_{\text{avg}}\Psi)_n\| &\le \frac{1}{4}\Big(\|\delta B_n\|\|\psi_{n+1} + \psi_n\| + \|\delta B_{n-1}\|\|\psi_n + \psi_{n-1}\|\Big) \\ &\le \frac{1}{4}\max_k \|\delta B_k\|\Big(\|\psi_{n+1}\| + 2\|\psi_n\| + \|\psi_{n-1}\|\Big). \end{aligned} \tag{A.4}$$

Squaring both sides and applying Cauchy–Schwarz yields:

$$\|(E_{\text{avg}}\Psi)_n\|^2 \leq \frac{1}{4}\left(\max_k \|\delta B_k\|\right)^2 \Big(\|\psi_{n+1}\|^2 + 2\|\psi_n\|^2 + \|\psi_{n-1}\|^2\Big). \tag{A.5}$$

Summing over all lattice sites $n = 1, \ldots, N$ to compute the global Hilbert space norm:

$$\begin{aligned}\|E_{\text{avg}}\Psi\|_{\mathcal{H}_N}^2 &= \sum_{n=1}^{N} \|(E_{\text{avg}}\Psi)_n\|^2 \\ &\leq \frac{1}{4}\left(\max_k \|\delta B_k\|\right)^2 \sum_{n=1}^{N}\Big(\|\psi_{n+1}\|^2 + 2\|\psi_n\|^2 + \|\psi_{n-1}\|^2\Big).\end{aligned} \tag{A.6}$$

Due to the periodic boundary conditions, shifting the summation indices reveals that $\sum \|\psi_{n+1}\|^2 = \sum \|\psi_{n-1}\|^2 = \sum \|\psi_n\|^2 = \|\Psi\|_{\mathcal{H}_N}^2$. Therefore:

$$\|E_{\text{avg}}\Psi\|_{\mathcal{H}_N}^2 \leq \frac{1}{4}\left(\max_k \|\delta B_k\|\right)^2 \Big(4\|\Psi\|_{\mathcal{H}_N}^2\Big) = \left(\max_k \|\delta B_k\|\right)^2 \|\Psi\|_{\mathcal{H}_N}^2. \tag{A.7}$$

Taking the square root and taking the supremum over $\|\Psi\|_{\mathcal{H}_N} = 1$ yields $\|E_{\text{avg}}\| \leq \max_n \|\delta B_n\|$, completing the proof with $C = 1$. □

Because E_{avg} scales linearly with the one-sided spatial gradient δB_n, it vanishes completely in a homogeneous obstruction field ($B_n \equiv B$). Furthermore, since its off-diagonal blocks match the order of spatial differences, its contribution to the global commutator structure $[H, H^\top]$ remains strictly bounded at order $\mathcal{O}(\varepsilon^2)$, leaving the linear $\mathcal{O}(\varepsilon)$ localized hotspot analysis in Section 7 unaffected.